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CLASS : DIV A

ROLL NO. : U24CS076

1. Do the installation of MinGW-Minimalist GNU for windows and show which of the C and C++ standard supported by the compiler installed in your system.

```
PS D:\oops> gcc --version
gcc.exe (MinGW.org GCC-6.3.0-1) 6.3.0
Copyright (C) 2016 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

PS D:\oops> g++ --version
g++.exe (MinGW.org GCC-6.3.0-1) 6.3.0
Copyright (C) 2016 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Write a basic program in C++ and execute it using command prompt. Write down the different commands you used during program writing and execution of the same.

```
PS D:\oops> cd lab-1
PS D:\oops\lab-1> g++ q2.cpp -o q2

PS D:\oops\lab-1> ./q2
Hello World
```

3. Write a program to read the input and print the output as mentioned below.

3.1. Input: 25

Output: 25

3.2. Input: 5.3267

Output: 5.3267

3.3. Input: 5.32 Output: 5.3200

3.4. Input: 5664.57567456

Output: 5664.57

3.5. Input: 35345364

Output: 35345364 * 35345364

3.6. Input & Output: Your name

3.7. Input & Output: S. V. National Institute of Technology, Surat

```
2
3 #include <bits/stdc++.h>
4 using namespace std;
5
6 Windsurf: Refactor | Explain | Generate Function Comment
7 int main()
8 {
9     int a;
10    cin >> a;
11    cout << a << endl;
12    double b;
13    cin >> b;
14    cout << b << endl;
15    double c;
16    cin >> c;
17    cout << fixed << setprecision(4) << c << endl;
18    double d;
19    cin >> d;
20    cout << fixed << setprecision(2) << d << endl;
21
22    string l;
23    cin >> l;
24    cout << l << "*" << l << endl;
25
26    getchar();
27
28    string s;
29    getline(cin, s);
30    cout << s << endl;
31
32    string t;
33    getline(cin, t);
34    cout << t << endl;
35
36    return 0;
37 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

5.3267
5.32
5.3200
5664.57567456
5664.57
35345364
35345364*35345364
rudraksh fause
rudraksh fause
S.V. National Institute of Technology, Surat
S.V. National Institute of Technology, Surat

4. Write a C++ program to compute the area of a circle.

```
3  #include <bits/stdc++.h>
4  using namespace std;
Windsurf: Refactor | Explain | Generate Function Comment
5  int main()
6  {
7      double r, a;
8      cin >> r;
9      a = 3.14 * r * r;
10     cout << a << endl;
11     return 0;
12 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\oops\lab-1> g++ q4.cpp -o q4
PS D:\oops\lab-1> ./q4
10
314
PS D:\oops\lab-1>

5. Write a C++ program to convert a given temperature in Fahrenheit to Celsius.

```
2
3 #include <bits/stdc++.h>
4 using namespace std;
Windsurf: Refactor | Explain | Generate Function Comment
5 int main() {
6     double f, c;
7     cin >> f;
8     c = (f - 32) * 5 / 9;
9     cout << c << endl;
10    return 0;
11 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\oops\lab-1> g++ q5.cpp -o q5
PS D:\oops\lab-1> ./q5
-40
-40
PS D:\oops\lab-1>

6. Write a C++ program to add, subtract, multiply and divide two integers given by the user.

```
2
3 #include <bits/stdc++.h>
4 using namespace std;
Windsurf: Refactor | Explain | Generate Function Comment
5 int main() {
6     int a,b;
7
8     cin >> a >> b;
9     cout << a + b << endl;
10    cout << a - b << endl;
11    cout << a * b << endl;
12    cout << a / b << endl;
13
14    return 0;
15 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + - [] [] []

```
PS D:\oops\lab-1> g++ q5.cpp -o q5
PS D:\oops\lab-1> ./q5
-40
-40
PS D:\oops\lab-1> g++ q6.cpp -o q6
PS D:\oops\lab-1> ./q6
3
5
8
-2
15
0
```

7. Write a C++ program to take length as an input and find the area and perimeter of a rectangle.

```
1
2
3 #include <bits/stdc++.h>
4 using namespace std;
Windsurf: Refactor | Explain | Generate Function Comment
5 int main() {
6     double l, a, p;
7     cin >> l;
8     a = l * l;
9     p = 2 * (l + l);
10    cout << a << endl;
11    cout << p << endl;
12    return 0;
13 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\oops\lab-1> g++ q7.cpp -o q7
PS D:\oops\lab-1> ./q7
10
100
40
PS D:\oops\lab-1>

8. Write a C++ program to swap values of two variables with and without using a third variable.

```
q8.cpp > ...  
4  
Windsurf: Refactor | Explain | Generate Function Comment  
5 int main() {  
6     int a, b, temp;  
7     cin >> a >> b;  
8     temp = a;  
9     a = b;  
10    b = temp;  
11    cout << a << " " << b << endl;  
12  
13    a = a + b;  
14    b = a - b;  
15    a = a - b;  
16    cout << a << " " << b << endl;  
17  
18    return 0;  
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\oops\lab-1> g++ q8.cpp -o q8  
PS D:\oops\lab-1> ./q8  
3  
5  
5 3  
3 5  
PS D:\oops\lab-1> 
```

9. Write a C++ program to calculate simple and compound interest.

```
q9.cpp > ...
1
2 #include <bits/stdc++.h>
3 using namespace std;
4
Windsurf: Refactor | Explain | Generate Function Comment
5 int main() {
6     double p, r, t, si, ci;
7     cin >> p >> r >> t;
8     si = p * r * t / 100;
9     ci = p * pow(1 + r / 100, t) - p;
10    cout << si << endl;
11    cout << ci << endl;
12    return 0;
13 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell + v [] ... [] x

```
PS D:\oops\lab-1> ./q9
10
10
10
10
15.9374
PS D:\oops\lab-1> 
```

10. Write a program to input two numbers and display the maximum number.

Windsurf: Refactor | Explain | Generate Function Comment

📄 powershell + ▾ 📄 🗑️ ... 📄 ✕

```
PS D:\oops\lab-1> ./q10
```

5

```
PS D:\oops\lab-1>
```

100